

VOLTANOR SUSPENSION 10MG/5ML

Composition:

Each 5 ml suspension contains Diclofenac Potassium 10mg

Description:

Off white to pale yellow coloured suspension having fruity taste

Pharmacodynamics:

Diclofenac potassium is a nonsteroidal anti-inflammatory drug with pronounced antirheumatic, anti-inflammatory, analgesic and antipyretic properties. It is an inhibitor of prostaglandin biosynthesis (prostaglandin is important in the pathogenesis of inflammation, pain and fever) and modulator of arachidonic acid release and uptake.

In rheumatic conditions, diclofenac is effective against chest pain, pain on movement, morning stiffness, swelling of the joints and functional performance.

Diclofenac potassium rapidly relieves spontaneous pain and pain on movement and reduces inflammatory swelling and wound oedema in post-traumatic and postoperative inflammatory conditions.

In migraine attacks, diclofenac is found to be effective in relieving the headache and in improving the accompanying symptom of nausea.

Diclofenac potassium has also shown marked analgesic effect in moderate and severe painful states of non-rheumatic origin.

In vitro, diclofenac potassium does not suppress proteoglycan biosynthesis in cartilage at concentrations equivalent to the concentrations reached in human beings.

Diclofenac potassium has a rapid onset of action and is therefore suitable for the treatment of acute episodes of pain and inflammation. It is also found to exert a pronounced analgesic effect in moderately and severely painful states.

Diclofenac potassium relieves the pain and reduces the extent of bleeding in primary dysmenorrhoea.

Pharmacokinetics:

Absorption: Diclofenac is rapidly and completely absorbed. Food intake does not affect extent of absorption, but time taken to reach C_{max} is longer.

Distribution: Peak plasma concentration after one 50mg tablet was 1.2 pg/mL (3.9 pmol/L) after 20-60 min. The plasma concentrations show a linear relationship to the size of the dose. Administration of 50mg 3 times daily for a week does not lead to cumulation of diclofenac in the plasma.

Diclofenac is highly bound to serum proteins (99.7%), mainly albumin (99.4%).

Diclofenac enters the synovial fluid, and peak concentrations are measured 2-4 hrs after peak plasma levels have been reached. The apparent elimination half-life from the synovial fluid is 3-6 hrs. This shows that the concentrations in the synovial fluid are already higher than plasma concentrations 4-6 hrs after administration, and they remain higher for up to 12 hrs.

Metabolism: Roughly half of diclofenac potassium undergoes first-pass metabolism; consequently, the areas under the concentration curves (AUCs) after an oral dose are only about half as large as after a parenteral administration of a same dose.

The biotransformation of diclofenac involves partly glucuronidation of the intact molecule but mainly single and multiple hydroxylation followed by glucuronidation.

Elimination: The total systemic clearance of diclofenac in plasma is 263 ± 56 mL/min (mean value $\pm$ SD). The terminal half-life in plasma is 1-2 hrs. About 60% of the dose is excreted via the kidneys in the form of metabolites; <1% is excreted as unchanged substance. The remainder is eliminated as metabolites via the bile in the faeces.

Elderly: No relevant age-dependent differences have been observed in absorption, metabolism and excretion.

Impaired renal function: No accumulation of the unchanged active substance can be inferred from the single-dose kinetics under the usual dosage schedule. In patients with a creatinine clearance of <10 mL/min, the theoretical steady-state plasma levels of metabolites are about 4 times higher than in healthy subjects. The metabolites are nevertheless ultimately cleared via the bile. Impaired hepatic function (chronic hepatitis, nondecompensated liver cirrhosis): The kinetics and metabolism of diclofenac are the same as in healthy patients.

Indication:

Short-term treatment in the following acute conditions: Posttraumatic and postoperative pain, inflammation and swelling eg, following dental or orthopaedic surgery; as an adjunct in acute painful inflammatory infections of the ear, nose or throat eg, pharyngotonsillitis and otitis. In keeping with standard therapeutic principles, the underlying disease should be treated with basic therapy, as appropriate.

Recommended dose:

After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

Children ≥ 1 year should be given 0.5-2mg/kg bodyweight daily, depending on the severity of the condition.

In adolescents ≥ 14 years, it is usually sufficient with a daily dose of 75-100mg.

The daily dosage should be prescribed in 2-3 fractional doses.

The maximum daily dose of 150mg should not be exceeded.

As a general recommendation, the dose should be individually adjusted. Adverse effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see section WARNINGS AND PRECAUTIONS).

ESTABLISHED CARDIOVASCULAR DISEASE OR SIGNIFICANT CARDIOVASCULAR RISK FACTORS.

Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤ 100 mg daily if treated for more than 4 weeks (see section WARNINGS AND PRECAUTIONS).

Route of administration:

Oral administration

Contraindication:

Peptic ulcer. Hypersensitivity to diclofenac. Similar to other nonsteroidal anti-inflammatory agents, it is contraindicated in patients whom attacks of asthma, urticaria or acute rhinitis are precipitated by acetylsalicylic acid or other drugs with prostaglandin synthetase-inhibiting activity.

Severe cardiac failure (see section WARNINGS AND PRECAUTIONS).

Warning and precautions:*Risk of GI Ulceration, Bleeding and Perforation with NSAID*

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Cardiovascular Thrombotic Events

Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose or duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially

in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.

There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAID use.

Hypertension

NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired anti-hypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.

Heart Failure

Fluid retention and oedema have been observed in some patients taking NSAIDs, therefore caution is advised in patients with fluid retention or heart failure.

Gastrointestinal Events

All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patients about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

Severe Skin Reactions

NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens - Johnson syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reactions and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.

Cardiovascular Effects

Treatment with NSAIDs including diclofenac, particularly at high dose and in long term, maybe associated with an increased risk of serious cardiovascular thrombotic events (including myocardial infarction and stroke).

Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤ 100 mg daily when treatment continues for more than 4 weeks.

As the cardiovascular risks of diclofenac may increase with dose and duration of exposure, the lowest effective daily dose should be used for the shortest duration possible. The patient's need for symptomatic relief and response to therapy should be re-evaluated periodically, especially when treatment continues for more than 4 weeks.

Patients should remain alert for the signs and symptoms of serious arteriothrombotic events (e.g. chest pain, shortness of breath, weakness, slurring of speech), which can occur without warnings. Patients should be instructed to see a physician immediately in case of such an event.

Precautions

Strict accuracy of diagnosis and close medical surveillance are imperative in patients with symptoms indicative of gastrointestinal disease, with a case history suggestive of gastrointestinal ulceration, with ulcerative colitis or with Crohn's disease, as well as in patients suffering from severe impairment of hepatic function. Gastrointestinal bleeding or ulcerations/perforations in elderly patients can occur at any time during the treatment, even without warning symptoms or a

previous history. In the rare instances where gastrointestinal ulcerations or bleeding occur in patients under treatment with diclofenac, the drug should be withdrawn.

Particular caution should be taken in patients with impaired cardiac or renal function, in the elderly, in patients being treated with diuretics, and in those with extracellular volume depletion from any cause, eg in the peri- or postoperative phase of major surgical operations. Monitoring of renal function as a precautionary measure is recommended. Discontinuation of the medication is followed by recovery to the pretreatment state. The lowest effective dosage is recommended in frail elderly patients or those with a low body weight.

As with other NSAIDs, elevations of one or more liver enzymes may occur. During prolonged treatment with diclofenac, monitoring of hepatic function is recommended. Withdraw the drug if impairment of liver function persists or worsens, if clinical signs or symptoms consistent with liver disease develop, or if other manifestations occur (eg eosinophilia, skin rash, etc). Use of diclofenac in patients with hepatic porphyria may trigger an attack.

Treatment with diclofenac in the aforementioned indications generally proves necessary only for a few days, but if diclofenac is administered over a more prolonged period, it is advisable, as with other highly active nonsteroidal anti-inflammatory agents, to perform blood counts.

As in the case of other NSAIDs, allergic reactions, including anaphylactic/anaphylactoid reactions, can occur even without earlier exposure to the drug. Discontinue diclofenac immediately if rash occurs.

Gastrointestinal effects

NSAIDs, including diclofenac, may be associated with increased risk of gastrointestinal anastomotic leak. Close medical surveillance and caution are recommended when using diclofenac after gastrointestinal surgery.

Interaction with other medicaments:

Diclofenac may raise their plasma concentrations when given with preparations containing lithium or digoxin.

Concomitant treatment with potassium-sparing diuretics may raise the serum potassium levels, thus making it necessary to monitor the latter. Concomitant systemic administration of different nonsteroidal anti-inflammatory agents or of glucocorticoids may increase the occurrence of adverse effects.

There have been isolated reports of an increased risk of haemorrhage during combined treatment with diclofenac and anticoagulants. Like other NSAIDs, diclofenac in high doses (200mg) may temporarily inhibit platelet aggregation.

Clinical studies have shown that diclofenac can be given together with oral antidiabetic agents without affecting their clinical effect. However, there have been isolated reports of both hypoglycaemic and hyperglycaemic reactions, which necessitated changes in the dosage of the antidiabetic agents.

Cases of nephrotoxicity have been reported when methotrexate and NSAIDs are given 24 hours of each other. This interaction is mediated through accumulation of methotrexate resulting from impairment of renal excretion in the presence of the NSAID.

Convulsions may occur due to an interaction between quinolones and NSAIDs. This may occur in patients with or without a history of epilepsy or convulsions.

Co-administration of diclofenac with aspirin or corticosteroids may increase the risk of gastrointestinal bleeding. Concomitant use of two or more NSAIDs should be avoided. Concomitant use of cardiac glycosides and NSAIDs in patients may exacerbate cardiac failure, reduce GFR and increase plasma glycoside levels.

As NSAIDs can reduce the effects of mifepristone, they should not be used for 8-12 days after mifepristone administration.

Concomitant use of NSAIDs with antihypertensive drugs (eg beta-blockers, angiotensin converting enzyme (ACE) inhibitors, diuretics) may cause a decrease in their antihypertensive effect via inhibition of vasodilatory prostaglandin synthesis.

Pregnancy and lactation:

Diclofenac should not be administered during pregnancy due to the possibility of premature closure of the ductus arteriosus or to that of uterine inertia.

Diclofenac is excreted in the breast milk.

Side effects:*Gastrointestinal Tract*

Occasional: Epigastric pain, other gastrointestinal disorders (e.g. nausea, vomiting, diarrhoea, abdominal cramps, dyspepsia, flatulence, anorexia).

Rare: Gastrointestinal bleeding, haematemesis, melena, peptic ulcer with or without bleeding or perforation, bloody diarrhoea.

Isolated cases: Lower gut disorders (e.g. nonspecific haemorrhagic colitis and exacerbation of ulcerative colitis or Crohn's disease), aphthous stomatitis, glossitis, oesophageal lesions, constipation.

Central (and Peripheral) Nervous System

Occasional: Headache, light-headedness, vertigo.

Rare: Drowsiness

Isolated cases: Disturbances of sensation including paraesthesia, memory disturbances, disorientation, disturbances of vision (reduced visual acuity, diplopia), impaired hearing, tinnitus, insomnia, irritability, convulsions, depression, anxiety, nightmares, tremor, psychotic reactions, taste alteration disorders.

Skin

Occasional: Rashes or skin eruptions.

Rare: Urticaria

Isolated cases: Bullous eruptions, eczema, erythema multiforme, Stevens-Johnson syndrome, Lyell's syndrome (acute toxic epidermolysis), erythroderma (exfoliative dermatitis), loss of hair, photosensitivity reactions, purpura (including allergic purpura).

Kidney

Isolated cases: Acute renal failure, haematuria, proteinuria, interstitial nephritis, nephrotic syndrome, papillary necrosis.

Liver

Occasional: Elevation of serum aminotransferases (SGOT, SGPT).

Rare: Hepatitis with or without jaundice.

Isolated cases: Fulminant hepatitis.

Blood

Isolated cases: Thrombocytopenia, leucopenia, agranulocytosis, haemolytic anaemia, aplastic anaemia.

Hypersensitivity

Rare: Hypersensitivity reactions, e.g. asthma, anaphylactic/anaphylactoid systemic reactions including hypotension.

Other Organ Systems

Rare: Oedema

Isolated cases: Impotence (association with ingestion of diclofenac doubtful), palpitation, chest pain, hypertension.

Cardiac disorders:

Kounis syndrome : Frequency "not known"

Uncommon*: Myocardial infarction, cardiac failure, palpitations, chest pain.

* The frequency reflects data from long-term treatment with a high dose (150 mg/day).

Description of Selected Adverse Drug Reactions*Arteriothrombotic events*

Meta-analysis and pharmacoepidemiological data point towards an increased risk of arteriothrombotic events (for example myocardial infarction) associated with the use of diclofenac, particularly at a high dose (150 mg daily) and during long-term treatment (see section WARNINGS AND PRECAUTIONS).

Overdosage:

Management of acute poisoning with NSAIDs essentially consists of supportive and symptomatic measures. There is no typical clinical picture resulting from an overdosage of diclofenac. Supportive and symptomatic treatment should be given for complications such as

hypotension, renal failure, convulsions, gastrointestinal irritation and respiratory depression. Specific therapies such as forced diuresis, dialysis or haemoperfusion are unlikely to be helpful in eliminating NSAIDs due to their high rate of protein binding and extensive metabolism.

Storage conditions:

Store below 30°C

Presentation:

60, 90, and 120ml in amber glass bottles

Manufacturer and Product Registration Holder:

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